

ESSABIR MOHAMED

El jadida, Morocco | mohamedessabir20@gmail.com | +212635930321 | [linkedin.com/in/mohamed-essabir](https://www.linkedin.com/in/mohamed-essabir)
github.com/mohamed2020m

Profile

I am a Computer Engineering and Emerging Technologies student at ENSA El Jadida, currently interning at Oracle. Passionate about full-stack development and emerging technologies, I strive to deepen my technical expertise and collaborate effectively in a professional environment. I am highly adaptable, detail-oriented, and eager to learn from experienced mentors while contributing meaningfully to team projects.

Education

- **Computer Engineering and Emerging Technologies**, ENSAJ - El Jadida - Morocco Sept 2021 - Present
- **Preparatory Cycle**, ENSAJ - El Jadida - Morocco Sept 2019 - June 2021
- **Baccalaureate in Mathematics**, Lycée Technique Ibn Soulayman Rasmouki - Tiznit - Morocco Sept 2018 - June 2019

Experience

Machine Learning Engineer, THE UM6P VANGUARD Center - Internship Jul 2024 – Sept 2024

- Developed QuizMasterAI, a question generation and quiz management system using fine-tuned LLM models to answer questions and generate quizzes.
- Designed a role-based application allowing students to take quizzes generated by professors, with a dashboard to manage quizzes, statistics, and users, as well as language and LLM model selection.

Backend Developer / ML, 3D SMART FACTORY - Internship Jul 2023 - Sept 2023

- Managed and assigned tasks within the team, ensuring project coordination and follow-up.
- Developed the backend of the application with FastAPI for model deployment and query management.
- Deployed the MeshSegNet model for tooth segmentation, ensuring inference and deployment on AWS.
- Collaborated in developing the frontend of the application, built with React, to integrate model results into an interactive user interface.

Full Stack Developer, ESEF El Jadida - Internship Jul 2022 - Sept 2022

- Developed and maintained web applications for the pre-registration of ESEF students using Laravel, PHP 8, and implemented responsive design compatible across browsers.
- Integrated secure authentication, protected sensitive information, and used Git for version control and code integrity.

Frontend Developer, Z SMART SERVICE - Internship Apr 2022 - Jun 2022

- Developed an interactive web application for tutorial tracking and hands-on learning, with intuitive and responsive user interfaces.
- Integrated dynamic features to enhance the user experience, including management of tutorial content and personalized learning paths.

Projects

TerraViva: AI-Enhanced 3D Learning Platform for Medical Education <https://shorturl.at/6u0hG>

- TerraViva is a groundbreaking educational platform that transforms medical education through immersive, AI-powered learning experiences. It integrates advanced technologies through Spring Boot and FastAPI backends, a Next.js frontend, and a Flask mobile app to provide a modern and interactive learning experience.
- Tools used: Git, PostgreSQL, Node.js, JDK, Flutter, Python, React, Next.js, NextAuth.js, TailwindCSS, Spring Boot, Spring Security, FastAPI

TeethSeg for 3D Tooth Segmentation with Deep Learning

teethseg.vercel.app

- TeethSeg is a web application dedicated to 3D tooth segmentation, built with Vite and ReactJS. It leverages the MeshSegNet deep learning model to provide precise and efficient segmentation features.
- Tools used: JavaScript, React.js, Tailwind, Three.js, VTK.js, Firebase, Python, FastApi, Pytorch, scikit-learn, AWS Lambda, AWS S3, AWS ECR, Docker.

Certify-Chain: Blockchain Solution for Certification Validation

certify-chain.vercel.app

- A next-generation platform based on blockchain for verifying degrees.
- Tools used: JavaScript, Next.js, React.js, MUI, Web3, Solidity, Blockchain, MetaMask, Ethereum, Truffle.

SlideController: Remote Control for PPTX Presentations

<https://shorturl.at/YFyWU>

- SlideController is a mobile application and server-based solution that allows users to remotely control PPTX slides on a PC. The system consists of an Android application (client side) and an RCPServer (server side) for seamless presentation control. Key features include slide navigation and QR code scanning for easy connection, eliminating the need to manually enter the IP address and port. Ideal for lecturers and presenters, it offers a simple and efficient way to control slides remotely.
- Tools used: Java, Android, Gradle, JavaFX.

Taper Angle

<https://shorturl.at/9nla9>

- An innovative application developed in Android (Java) integrating OpenCV to calculate the taper angle, specifically used for dental analyses.
- Tools used: Java, Android.

EventJ

fablab-ucd.org

- In response to the FABLAB event organized by ENSAJ in 2022, I created a user-friendly web page that served as a comprehensive resource hub. This platform contained all the essential information needed for participants, ensuring smooth access to event details and updates.
- Tools used: HTML5, CSS3, JavaScript, JQuery

Hackathons

InfinityChat: Hackathon AI Chatbot for Regulations - Lablab.ai

infinity-chat.vercel.app

- InfinityChat is a hackathon project aimed at building a chatbot capable of answering questions and providing information about regulations and procedures in various fields, such as education, healthcare, and banking systems in Saudi Arabia.

Forest Rangers - GAIATHon

- Participated in GAIATHon'24 with the Ni'ma project, an innovative technological solution using AI and IoT for forest protection, which won the best preliminary project award.

SmartBoard Interactive Display - ENSAJ

- Participated in a hackathon on modernizing educational and administrative processes at ENSAJ, proposing an innovative solution integrating interactive smart boards. Our project won second place due to its significant impact on efficiency and sustainability.

Technologies

Languages: Python, JavaScript/TypeScript, Java, Kotlin, Dart, C#, PHP, SQL

Frontend Frameworks/Libraries: React.js, Next.js, Angular

Backend Frameworks: Django, FastAPI, Flask, Spring/Spring Boot, .NET, Laravel, Node.js, Express

Mobile Development: Android, Flutter, React Native

Databases: Microsoft SQL Server, PostgreSQL, MySQL, MongoDB

Cloud & Infrastructure: AWS, Terraform

Containerization & Orchestration: Docker, Kubernetes

DevOps & Automation: CI/CD pipelines

Version Control & Collaboration: Git, GitHub, GitLab

Project Management Tools Trello, Jira

Languages

- English (Professional proficiency)
- Français (Courant)
- Moroccan Darija/Arabic (Native)

VOLUNTEER WORK

SHIP-Club (Solidary Humans Involved in Peace) - ENSAJ

[ship-club on LinkedIn](#)

is a humanitarian and associative club based at the École Nationale des Sciences Appliquées d'El Jadida (ENSAJ). The club is dedicated to promoting solidarity, peace, and social responsibility among students and the local community through initiatives such as organizing charity events, hosting educational and cultural workshops, and encouraging volunteer work and community development. As a member of SHIP-Club for two years, I actively contributed to various impactful projects, fostering collaboration and mutual respect among peers while developing my interpersonal and leadership skills. This experience strengthened my sense of empathy and responsibility, enabling me to create positive social impacts in both academic and local environments.